

K-Bound on Office-Home — Real-Shift Benefit-Sign Benchmark

Dataset: Office-Home (4 domains, 65 classes, 15,588 images) · **Shift:** covariate (Real-World → Art / Clipart / Product) · **Date:** 2026-06-15 · 6th real-shift candidate for the K-Bound paper

VERDICT — NULL. beats_both = FALSE on both the val regime scan and the held-out test.

Office-Home is a **safety / abstention** case, not a beats-both win. The best deployed adapter is net-helpful, and the only genuinely *mixed* domain (Product) has harm that is **not label-free detectable on source→target transfer**. The "goldilocks" precondition (a domain that is mixed *and* detectable for the deployed adapter) is structurally absent: detectability and mixedness are anti-correlated across domains.

1. Setup & protocol (integrity-clean)

- **Source model f0:** ResNet-50 (ImageNet-V2) fine-tuned on Real-World **train** → **86.1%** Real-World-val accuracy. Fixed 8-epoch budget (no early-stopping on the calibration split).
- **14 adaptation candidates:** Tent / EATA / SAR × {online, episodic} × {mild, aggressive} (12) **plus** two new inference-only candidates — **label-shift** (MLLS target-prior logit correction, no parameter update) and **conservative** (damped + clipped prior correction, no parameter update).
- **Conditions:** 2 seeds × {iid, imbalanced, single-class stream} × {tiny=8, small=16} batch regimes; eval pool = 320 class-balanced images/condition. Source 12 conditions; target-val 36 (504 records); target-test 36 (504 records).
- **Held-out discipline:** τ^* , the candidate set, the deployed adapter, and the conformal certificate are calibrated on **source (Real-World/val) + target-VAL only**. The test split is touched **once**. No tuning on test. The whole panel is reported, including all nulls.
- **Benefit B** = acc(adapted) – acc(frozen), recomputed audit-grade from stored predictions.

2. The decisive question — gradient-TTA detectability

Can the label-free evidence detect the *subtle* Tent/EATA/SAR harm (not the trivially-detectable label-shift collapse, which is excluded here)? A feature separates it **in-sample**, but the **source-calibrated certificate does not transfer** to the target.



domain	regime (val / test)	% harmful (val / test)	best single-feature harm-AUC, in-sample (val / test)	certificate transfer-AUC, source→target (val / test)
Art	helpful-dom / helpful-dom	9% / 8%	0.90 / 0.99	0.71 / 0.63
Clipart	helpful-dom / helpful-dom	3% / 7%	0.99 / 0.94	0.85 / 0.82

Product	mixed / mixed	39% / 23%	0.69 / 0.80	0.53 / 0.63
Pooled (n≈432)	mixed / helpful-dom	17% / 12%	0.81 / 0.79	0.33 / 0.49 (anti-transfers)

Reading: Art and Clipart have cleanly *detectable* harm — but they are helpful-dominated, so there is almost no harm to avoid. Product is the only genuinely *mixed* domain, and there the harm is **not** label-free detectable (transfer-AUC $\approx$ chance). No domain sits in the goldilocks corner. The pooled certificate transfer-AUC is **below chance**, the same source→OOD non-transfer that sank the iWildCam case.

3. Router verdict — K-Bound does not beat both

Deployed (source-best) adapter = **eata_online_mild**, which is **net-helpful**, so the safety/freeze play cannot beat always-adapt.

route	metric	VAL	HELD-OUT TEST
route-a (deployed single candidate)	KGA regret	0.0281	0.0269
	always-freeze regret	0.0281	0.0269
	always-adapt regret	0.0009	0.0000
	beats_both / false-adapts	False / 0	False / 0
route-b (multi-candidate, $\tau^*=1.12$ source-cal)	router regret	0.0347	0.0324
	always-freeze regret	0.0347	0.0324
	best-fixed-adapt regret	0.0056	0.0054
	decisions (A / F / Abstain)	0 / 2 / 34	0 / 8 / 28
	beats_both	False	False

K-Bound makes **0 false-adapts** (safe — it ties always-freeze), but it loses to always-adapt and best-fixed-adapt. route-b abstains on almost everything because the source-calibrated τ^* does not transfer to the target scale.



4. Full candidate panel (target-val, pooled over domains)



candidate	mean B	% harmful	candidate	mean B	% harmful
eata_online_mild (<i>deployed</i>)	+0.027	6%	sar_online_aggressive	−0.042	36%
tent_online_mild	+0.029	6%	tent_online_aggressive	+0.009	28%
eata_online_aggressive	+0.026	6%	eata_episodic_mild	+0.018	22%
sar_online_mild	+0.024	11%	tent_episodic_mild	+0.019	19%
tent_episodic_aggressive	+0.020	17%	sar_episodic_aggressive	+0.019	19%
sar_episodic_mild	+0.018	17%	eata_episodic_aggressive	+0.019	17%
conservative (<i>new</i>)	−0.012	81%	label-shift (<i>new</i>)	−0.466	100%

The two new candidates behaved as designed but did not rescue a win: **label-shift** is catastrophic (and trivially detectable → excluded from the decisive AUC), and **conservative** turned out **mildly harmful, not helpful** — so the router has no safe-helpful option to route toward. The only genuinely mixed gradient candidate is `sar_online_aggressive` (−0.042, 36% harm).

5. Interpretation — why it's a null

A real-shift beats-both needs a deployed adapter that itself **flips helpful↔harmful detectably**. On Office-Home: (i) the source-best adapter is **helpful-leaning** (12% harm on test, mean B +0.015) so abstaining/freezing is the wrong default; (ii) the harm that does exist is concentrated in Product, where it is **label-free undetectable on transfer**; (iii) the certificate anti-transfers (pooled 0.33 val / 0.49 test) and τ^* does not transfer ($\approx$ full abstention). This is the **same structural obstruction** as the prior five real-shift nulls (Camelyon17, RxRx1, ImageNet-R, ACDC, iWildCam) — now shown clearly **within a single covariate-shift dataset**, the case that was the best a-priori bet for label-free detectability.

This **reinforces** the paper's honest characterization: K-Bound is **decisive on synthetic corruption** (CIFAR-10-C, ImageNet-C/SAR), and a **safety/abstention method on real shifts** (does no harm, never beats both).

6. Reproducibility & integrity

- **Code:** `experiments/kbound/officehome/` — `oh_data.py` , `oh_candidates.py` , `oh_analyze.py` , `run_officehome_kbound.py` , `oh_report.py` , `supervise_oh.sh` .
- **Manifests:** `results/officehome_full_{source,targetval,targettest}/` (every number traces to `records[]/conditions[]`); consolidated in `results/officehome_full_FINAL/` (FINAL_SUMMARY.json, VERDICT_val.json, VERDICT_test.json, REGIME_SCAN.json, panel.csv, this PDF + figures).
- **f0:** `results/officehome_f0/f0_resnet50_rw_seed0.pt` . **Locked splits:** `results/officehome_kbound/splits.json` . **Data:** `~/kbound_officehome` (internal) + `experiments/kbound/data/office_home` (T9).
- **Integrity notes:** nothing tuned on test; τ /candidates fixed on val before test; held-out test evaluated once. One MPS stall during the test sweep was traced to GPU contention from a concurrent `run_iwildcam_kbound` job (not heat); after killing it the test completed a clean 36/36. Null reported straight.